

2	(a) obeys the equation $pV = nRT$ or $pV/T = \text{constant}$ all symbols explained; T in kelvin/thermodynamic temperature	M1	A1	[2]
(b)	(i) temperature rise = 48 K	A1	[1]	
	(ii) $\langle c^2 \rangle \propto T$ or equivalent $\langle c^2 \rangle = (353/305) \times 1.9 \times 10^6$ $c_{\text{r.m.s.}} = 1480 \text{ m s}^{-1}$	C1	C1	A1
				[3]